

Transformer-based local-global guidance for image captioning

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ARTICLE INFO

Keywords:

Attention
Transformer
Encoder-decoder
Image captioning
Deep learning

ABSTRACT

Image captioning is a difficult problem for machine learning algorithms to compress huge amounts of images into descriptive languages. The recurrent models are popularly used as the decoder to extract the caption with significant performance, while these models have complicated and inherently sequential overtime issues. Recently, transformers provide modeling long dependencies and support parallel processing of sequences compared to recurrent models. However, recent transformer-based models assign attention weights to all candidate vectors based on the assumption that all vectors are relevant and ignore the intra-object relationships. Besides, the complex relationships between key and query vectors cannot be provided using a single attention mechanism. In this paper, a new transformer-based image captioning structure without recurrence and convolution is proposed to address these issues. To this end, a generator network and a selector network to generate textual descriptions collaboratively are designed. Our work contains three main steps: (1) Design a transformer-based generator network as word-level guidance to generate next words based on the current state. (2) Train a latent space to learn the mapping of captions and images into the same embedding space to learn the text-image relation. (3) Design a selector network as sentence-level guidance to evaluate next words by assigning fitness scores to the partial captions through the embedding space. Compared with the architecture of existing methods, the proposed approach contains an attention mechanism without the dependencies of time. It executes each state to select the next best word using local-global guidance. In addition, the proposed model maintains dependencies between the sequences, and can be trained in parallel. Several experiments on the COCO and Flickr datasets demonstrate that the proposed approach can outperform various state-of-the-art models over well-known evaluation measures.

1. Introduction

The topic of automatically extracting sentences for images has stimulated attentiveness in computer vision and language modeling analysis in recent years (Zhang, Li, Wang, Zhao, & Wang, 2021). Recently, pictures are one of the most usable information on the web, and the demand for labeling, captioning, and annotating them has significantly increased.

Image captioning is an essential work that requires a semantic commentary over the image and the ability to produce an accurate caption (Wang, Wan, & Chan, 2022). The models employed for image description can be practically classified into three types: (1) Template-Based (TB) models; (2) Retrieval-Based (RB) models; and (3) Artificial Neural Network (ANN)-based models (Liu, Hu, Li, Yu, & Guan, 2022; Zhou, Zhang, Jiang, Zhang, & Fan, 2020).

The TB and RB models are the traditional models that have certain limitations (Horanyi et al., 2022). The TB models, which utilize a

constrained caption structure to produce image descriptions, are more difficult without diversity; and the RB models, which exploit captions available from the searched annotated images, cannot produce new descriptions (Bai & An, 2018).

Most of TB and RB methods require sufficient training samples to extract image descriptions. To consider this limitation, modern data-driven methods (Tao, Cheng, Qiu, & Stojanovic, 2022) can be used by deep learning techniques. Besides, to reduce the complexity of TB and RB methods, online reinforcement learning models can be used based on the subsystems transformation technique (Xin et al., 2022). Furthermore, the presence of outliers decrease the ability of TB and RB models. To this end, stochastic nonlinear models (Stojanovic & Nedic, 2016) can be employed relying on accurate knowledge of the model and its parameters.

Differently, the ANN-based models make image sentences utilizing an encoder-decoder architecture, which is generally employed in image captioning. First, the image is encoded into a visual vector by an

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encoder. Then, this vector is forwarded into a decoder to extract caption words that can describe the image content (Huang, Chen, Ouyang, Wan, & Xue, 2020; Kulkarni et al., 2013). The popular encoders utilize Convolutional Neural Networks (CNNs) (Bai & An, 2018; Wang, Shen, & Ji, 2022), which are appropriate for feature extraction in image processing (Ren, He, Girshick, & Sun, 2015). Besides, the decoders generally utilize the Long-Short-Term-Memory (LSTM) as the decoder to extract textual information, and this structure has significant performance. (Horanyi et al., 2022; Lim, Chan, Ng, Fan, & Yang, 2022; Ren, Wang, Zhang, Lv, & Li, 2017; Tan, Chan, & Chuah, 2022; Zhu, Li, Liu, Li, et al., 2018). However, most existing ANN-based models focus on global content in the image and describe it with a similar caption. Hence, the extracted caption often describes global image content (Wang et al., 2022).

The LSTM can retain dependencies, while the LSTM architecture has sequential and complicated issues over time (Graves & Schmidhuber, 2005). In contrast to LSTM-based approaches, the transformer (Vaswani et al., 2017) structure contains the stacked attention module, avoiding recurrence, which doesn't depend on the previous time output in the learning process. Different from recurrent networks that can only attend to short-term relationships, transformers can attend to entire sequences by learning long-range relations. In addition, the straightforward structure of transformers provides processing multiple modalities utilizing similar processing modules and illustrates great scalability to large datasets and large capacity networks. Besides, the transformer has Multi-Head Attention (MHA) and Self-Attention (SA) components. The SA can adjust various positions and create a representation for all sequences, while the MHA can adjust various multi-modal representations for associating the caption and the image (Wang et al., 2022; Zhu, Li, Liu, Peng, & Niu, 2018).

Following previous transformer-based methods (Lin, Wang, Liu, & Qiu, 2021; Wang et al., 2022; Wei, Wu, Li, & Shi, 2022; Xu et al., 2015; Yu, Li, Yu, & Huang, 2019; Zhang, Shi, Mi, & Yang, 2021), attention can assist the model in attending to the essential regions in the image. However, recent transformer-based models allocate attention weights to each candidate vector based on the assumption that these vectors are relevant. In addition, current transformers cannot provide the intra-object relationships, while only consider the inter-object attention distribution. Besides, a single Attention Mechanism (AM) may not provide more complex relationships between key and query vectors.

To consider these limitations, we introduce a new transformer attention module without recurrence and convolution to the image description task. Our approach contains attention blocks without the dependencies of time, which retains dependencies between the sequences. Our method has three key parts. In the first step, we present a transformer-based generator network as a local policy to generate next words using the current state. Then, in the second step, we train a latent space to transfer captions and images into the same embedding space to learn the text-image relation. Finally, in the third step, we design a selector module as a global policy to select the next best word by assigning fitness scores to the partial captions through the embedding space. The contributions of our approach are as follows:

1) We present a transformer-based generator network as word-level guidance for predicting next words based on multi-modal information. The proposed generator architecture utilizes the transformer as a feature enhancer in the encoder and prepares appropriate information for the decoder.

Unlike previous works, such as (Graves & Schmidhuber, 2005; Li, Bao, Li, & Zhao, 2020; Wang et al., 2022; Xiao, Wang, Ding, Xiang, & Pan, 2019; Zhang et al., 2021; Zhong et al., 2021; Zhu, Li, Liu, Li, et al., 2018), which use an LSTM-based decoder, the proposed generator network indicates better efficiency to an LSTM-based structure on various measures. Moreover, we design a pre-layer normalization transformer to simplify the transformer structure for image feature enhancement. Therefore, our model is faster than the recurrent structures to learn relationships among feature vectors efficiently.

2) We design a selector module as sentence-level guidance, which

assigns fitness values to the partial captions to select the next best word.

Unlike recent models, such as (Ji, Du, & Zhang, 2021; Liu, Chen, Guo, Zhu, & Liu, 2021; Zha, Liu, Zhang, Zhang, & Wu, 2022; Zhang et al., 2021; Zhang et al., 2021; Zhu, Li, Liu, Peng, et al., 2018), which utilize only a local strategy to predict next words, the proposed selector module is employed as a global policy for word selection.

3) To learn the text-image relation, we create an embedding space, which learns the similarity between sentences and images, and can measure the correctness of extracted sentences. Therefore, our selector network assigns fitness scores to the partial caption through the embedding space.

4) We propose a Mask Attention Weight (MAW) module to constrain SA to attend on the most appropriate information.

Most existing transformer-based models, such as (Li, Yao, Guo, & Zhang, 2019; Wang et al., 2022; Wei et al., 2022; Xiong et al., 2020; Zhang et al., 2021; Zhu, Li, Liu, Peng, et al., 2018), assign attention weights to all candidate vectors based on the assumption that all vectors are relevant. It may generate some misleading or unrelated information. Therefore, the MAW module directly constrains the attention weights.

5) We design a Self-Eliminator Module (SEM) to filter out some irrelevant information in the object features.

Compared to the other transformer-based models, such as (Liu et al., 2021; Wan et al., 2022; Wang et al., 2022; Wei et al., 2022; Yu et al., 2019; Zhang et al., 2021; Zhu, Li, Liu, Peng, et al., 2018), that focus on the inter-object relations and ignore the intra-object relationships, the SEM considers the intra-object AM to eliminate the unrelated information and noise in feature vectors.

6) A single attention structure in transformers is unable to indicate more complex relationships between key and query vectors (Khan et al., 2021). Therefore, some appropriate relationships can be lost. To tackle this problem, we introduce a Multiple Attention Distribution (MAD) module to avoid a single shallow attention distribution and consider deeper internal relationships.

7) Extensive evaluations over the MSCOCO and Flickr30k datasets indicate the effectiveness of our framework.

2. Related works

In this section, we discuss the previous image captioning frameworks. As shown in Fig. 1, these models can be generally classified into three groups: Template-Based (TB), Retrieval-Based (RB), and ANN-based approaches (Bai & An, 2018; Xu et al., 2015; Yang, Wang, Chu, & Yang, 2022). Additional details of each model are described in their corresponding parts.

2.1. Template-Based models

TB models are traditional image description models, which require predefining an appropriate caption structure (Liu, Xu, & Wang, 2019). In TB models, sentences are produced via semantically and syntactically processes. First, the various actions, features, and attributes of the objects are extracted, and then these detections are used as templates for image description (Wang et al., 2022).

Due to the previous TB models ignoring the relationship between sentences and images (Stefanini et al., 2022), authors in (Farhadi et al., 2010) introduced an approach based on a score function for image description. This model is symmetric, where the best caption can be generated using a large dataset for the query image. However, grammatically correct captions cannot be produced by this method.

Therefore, in (Yang, Teo, Daumé III, & Aloimonos, 2011) a computationally practical model to consider these limitations was introduced, as well as the semantic content acquired based on a large semantically corpus. The Hidden Markov Algorithm (HMA) was employed as a caption generator (Yang et al., 2011). Although the captions produced by this model are both relevant and readable for query images, there are several defects in the extracted verbs and nouns, since the recognized

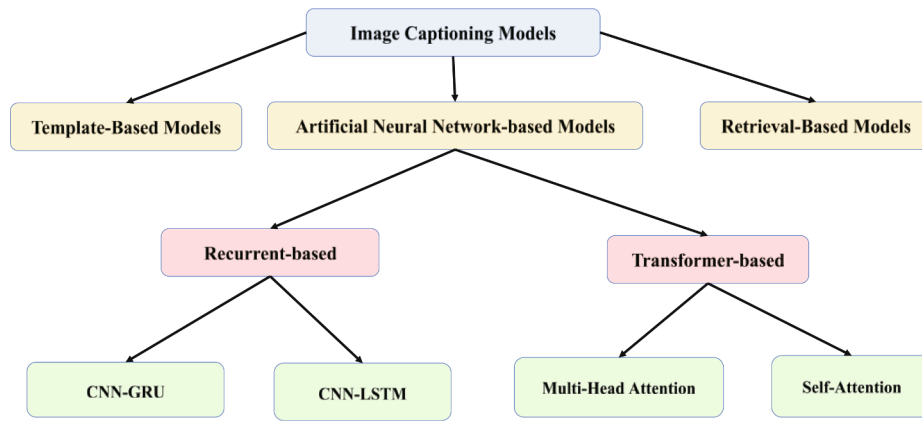


Fig. 1. An overview of image captioning models.

parts may be mistaken.

A model to utilize a caption template was developed in (Yang et al., 2011), where a quadruplet (Verbs-Prepositions-Scenes-Nouns) was used as a caption template. First, the authors use object detection frameworks (Felzenszwalb, Girshick, McAllester, & Ramanan, 2009) to detect scenes and objects. Then, the best quadruplet was generated based on the HMA structure, and the caption generation was obtained by providing the caption structure according to the quadruplet.

Authors in Li, Kulkarni, Berg, Berg, and Choi (2011) used semantic information, including attributes and objects to define a triplet for encoding recognition outputs to act as the sentence of the target image. Authors in Mitchell et al. (2012) represented the image by utilizing spatial relationships, actions, and objects as a tree-extracting process by using the recognition outputs. Authors in Kulkarni et al. (2013) proposed a compact relationship between the sentence and the image concept utilizing the big semantic data and the recognition models to detect objects in an image. This model can extract more relevant captions for images than other models.

In Kulkarni et al. (2013), a conditional structure was introduced to extract image contents according to attributes, objects, and spatial relationships in the query images. However, caption generation under the TB model is limited to image contents detected by visual methods, there are often issues with complexity, creativity, and coverage of extracted sentences. In addition, unlike human-written descriptions, utilizing constrained templates as the primary module of captions will make produced captions less natural (Deorukhkar & Ket, 2021).

2.2. Retrieval-based models

The RB models exploit descriptions available from the searched captioned images. Due to the generated captions of most RB models lacking creativity. In Farhadi et al. (2010), a (scene, action, object) latent space to join captions and images by solving an HMA was introduced. They proposed a novel measure to compute the semantic similarity between caption and image.

In addition, in Ordonez, Kulkarni, and Berg (2011) global descriptors were employed to retrieve some images from a web-scale group of described images. They employed semantic concepts of the retrieved information as final descriptions.

A data-driven model was designed in Kuznetsova, Ordonez, Berg, Berg, and Choi (2012), to exploit the image and captions. This model retrieved available human-written captions based on the similarity scores using linear programming. However, the generated descriptions may result in some mistakes.

After that, authors in Hodosh, Young, and Hockenmaier (2013) framed caption generation as a ranking task and utilized the canonical correlation to transfer the caption and image into a latent space. In addition, cosine similarity between caption and image was computed to

choose top-ranked captions for the target images.

To reduce the effects of noisy contents in models that rely on information retrieval for captioning, authors in Mason and Charniak (2014) employed a similarity function to retrieve a set of sentences based on the Word Probability Density (WPD) on the target image. The WPD was employed to consider the available sentences to extract the one with the highest score as a final description of the target image. In addition, the authors developed a non-parametric density estimation model to score the existing sentences.

Besides, to score the relationship between captions and images, two developments in caption retrieval were proposed in Ordonez et al. (2016), including retrieving the whole available images, and retrieving captions using the geometric distance of scene and object. Although the TB methods and RB models can produce relevant captions of images, these models require various predefined templates and cannot extract new sentences for a query image (Luo, Cheng, Jing, Zhao, & Song, 2022).

2.3. ANN-Based models

The ANN-based models make image sentences utilizing an encoder-decoder architecture (Liu et al., 2019). These models encode the image to a high-level representation and then decode this representation utilizing a language modeling algorithm. Current ANN-based models can be classified into two groups: (1) Recurrent-based models; and (2) transformer-based models. Some popular methods for each group are provided in their corresponding section.

2.3.1. Recurrent-based models

The recurrent-based encoder-decoder frameworks generally utilize a CNN encoder to transform the image into a vectorial representation, and an LSTM decoder to extract the corresponding outputs from the encoder representation. For example, authors in (Jia, Gavves, Fernando, & Tuytelaars, 2015) introduced a new version of the LSTM, that the semantic content was extracted at each time frame to guide the decoder in predicting the relevant caption to the image concept.

There are different recurrent-based methods, which employ the Attention Mechanism (AM) to determine which region of the target image is important (Graves & Schmidhuber, 2005; Wang et al., 2022; Zhu, Li, Liu, Li, et al., 2018). The AM enhances performance by forcing the algorithm to attend on the critical regions of the image when producing output captions. In Xu et al. (2015), two “soft” and “hard” AMs were employed to encode-decoder structure to better exploit the visual concept of an image. Most existing attention-based models utilize “soft” AM, because it can be embedded into a decoder for training directly. Hence, authors in Zhou, Xu, Koch, and Corso (2016) introduced a new text-conditional AM, which can interpret visual features using textual concepts.

A model was developed in Pedersoli, Lucas, Schmid, and Verbeek (2017) to consider the issue that previous AMs use image areas only to the LSTM state. In addition, a sequence-to-sequence LSTM structure was presented Liu, Sun, Wang, Wang, and Yuille (2017) to make a sequence of recognized regions to produce corresponding final descriptions.

To consider issues related to region misjudgment and object missing, in Li, Tang, Zhang, Deng, and Tian (2017), authors introduced a local-global AM, in which the object-level and image-level attributes were utilized by an AM. Furthermore, in Ye, Han, and Liu (2018), an LSTM was trained to solve the limitation that the detected object of the image includes irrelevant and relevant information.

There is the issue that basic AMs utilize the visual content, which can lead to cost raised direction with time progressing. To address this limitation in (Zhu, Li, Liu, Li, et al., 2018), a parallel LSTM structure was designed for image description, which can employ more parameters to achieve comparable performance. Moreover, in Luo, Price, Cohen, and Shakhnarovich (2018), a retrieval-based approach and a sentence extractor were used in a collaborative architecture.

To comprehend an image, some modern recurrent methods were developed to grasp many attributes of the objects and their similarity to represent the image at a higher level (Felzenszwalb et al., 2009). For instance, in Yang et al. (2019), a novel multitask model was proposed for cross-domain captioning that optimizes two objectives using a dual learning model.

Furthermore, Huang et al. (2020), an end-to-end trainable structure was designed to grasp attributes at each time frame by utilizing the word embedding of attributes. In Yang et al. (2020), authors improved an ensemble of RB and TB image description methods using a new dual generative adversarial structure.

In Wu et al. (2021), a local-global discriminative model was designed to generate local descriptive sentences. Besides, in Yu, Zhang, and Wu (2022) dual AM on pyramid visual feature maps was applied to consider the semantic-visual similarity to enhance the accuracy of extracted captions (Liu et al. (2022)). In Zha et al. (2022) a new LSTM-based policy structure for local caption generation was introduced, which employs visual attention as context for the current word.

However, recurrent-based structures need the information flow to be processed sequentially from the previous hidden states to the next one. This eliminates the possibility of utilizing parallelization and acceleration during training, and thus prevents the potential of recurrent-based architectures to make larger structures or process longer sequences (Khan et al., 2021).

2.3.2. Transformer-based models

The transformer-based models contain the stacked attention module, avoiding recurrence, which doesn't depend on the previous time output in the learning process (Khan et al., 2021). Different from recurrent networks that can only attend to short-term relationships, transformers can attend to entire sequences by learning long-range relations. Additionally, the straightforward structure of transformers provides processing multiple modalities utilizing similar processing modules and illustrates great scalability to large datasets and large capacity networks (Lin et al., 2021).

For instance, to produce the current word, the transformer only requires the previous Ground Truth (GT) word, and we don't require the previous state, which is utilized in the LSTM. Without recurrence, the transformer improves model training speed and extracts global dependencies among various inputs. Hence, recent models have shown the benefits of the transformer for image description (Vaswani et al., 2017; Yu et al., 2019; Zhu, Li, Liu, Peng, et al., 2018).

The encoder-decoder structure of the well-known transformer method (Guo et al., 2020) for image description is demonstrated in Fig. 2. It includes a decoder and an encoder with several transformer modules. The encoder generates encodings of visual information, while the decoder considers each of the encodings along with their contextual inputs to produce the output sequence. In addition, the transformer has

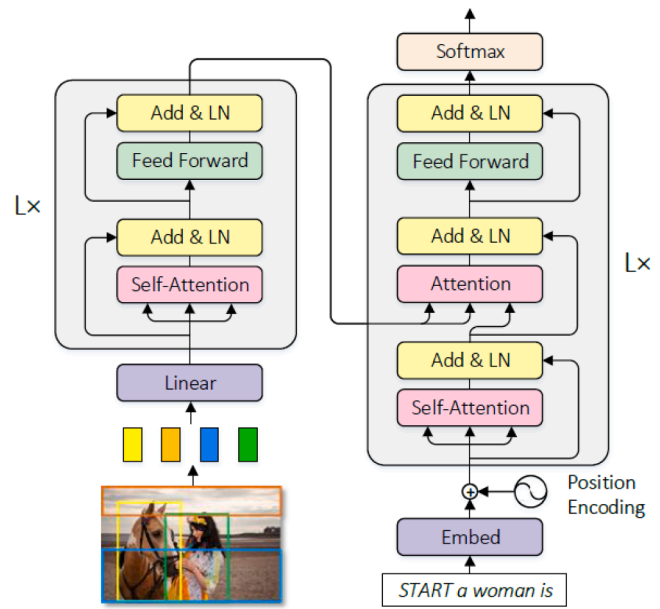


Fig. 2. Architecture of the well-known transformer method (Guo et al., 2020) for image captioning.

MHA and SA blocks. The SA can adjust various positions and create a representation for all sequences, while the MHA can adjust various multi-modal representations and associate the caption and the image. Besides, residual connections alongside layer normalization (Lin et al., 2021) are used after each block.

Inspired by the great success of transformer models in computer vision, researchers have recently employed transformer to solve image captioning issues. For example, in Zhu, Li, Liu, Peng, et al. (2018), a captioning transformer was introduced, which contains attention blocks without the dependencies of the time. They designed multilevel guidance to achieve better performance. Authors in Li et al. (2019), proposed a new transformer structure with 2 AMs to acquire the visual contents. They utilized the visual information to model internal similarity between the sequences as SA.

In addition, in Yu et al. (2019) the captioning transformer model was developed into a multi-modal space that simultaneously captures model interactions using an integrated attention module. In Guo et al. (2020) a new normalization structure was introduced to utilize it on the hidden activations within SA. In addition, a class of geometry attention was presented to efficiently exploit the important geometry relationships across the regions in the image.

Authors in Liu et al. (2021), presented the image description task from a novel sequence-to-sequence extraction perspective. They forwarded the visual features into a transformer structure to utilize global context at each encoder layer from the beginning and was convolution-free.

After that, in Wang et al. (2022) a geometry-aware transformer was proposed to obtain geometric representation ability for the encoder and decoder. This model explicitly employed spatial features in the image representation in encoding parts, and informed the decoder of word positions for producing captions.

However, recent transformer-based models allocate attention weights to every candidate vector, assuming that each vector is relevant and ignore the intra-object relationships. Besides, a single AM can't provide more complex intrinsic relationships between key and query vectors.

To consider these problems, we present a new transformer-based architecture without recurrence and convolutions to the image description task. To this end, we present a generator network and a selector module to collaboratively generate textual descriptions. Our

framework contains attention blocks without the dependencies of time. In the following, our architecture is predated in detail.

3. Proposed method

In this section, we elaborate the overall structure of our approach for image captioning. Our framework addresses image captioning limitations by exploiting word-level (local) and sentence-level (global) policies. Hence, we design a transformer-based generator network to generate next words in each state, and a selector network to compute the distance between the partial captions and visual features to select the best word.

Our method has three key parts. In the first step, we design a transformer-based generator network as local guidance to generate next words. Then, in the second step, we train a latent space to project captions and images into the same embedding space to learn the sentence-image similarity. Finally, in the third step, we design a selector network as global guidance to select the next best word by assigning fitness scores to the partial captions through the embedding space. The overall illustration of our model is provided in Fig. 3. Additional details of each part are presented in their corresponding section.

3.1. Transformer-based generator network

In this section, we employ the transformer-based AM to construct our multi-modal generator network. In the task of image captioning, our goal is to predict a caption $C = \{w_1, w_2, \dots, w_L\}$ for an image I , which precisely depicts the image information, where w_k denotes a word in caption C and L is the number of words. As shown in Fig. 4, the generator network engenders the probability to predict the next word at each state.

The generator network includes a visual encoder and a sentence decoder consisting of N layers. The encoder is the visual encoder, and the decoder is auto-regressive, employing the previously extracted words as additional information when producing the caption. Our structure contains several attention sub-layers. Following transformers, we apply sinusoidal “positional encoding” to the inputs at the bottoms of the decoder. The positional encoding features are of the same dimension as the input embedding, and are produced utilizing cosine and sine

equations of various frequencies (Ye et al., 2018). Due to the areas in the image do not have a natural order like sequences, no position encoding is used in the encoder part.

As shown in Fig. 4, the encoder inputs are the region vectors generated from FasterRCNN (Ren et al., 2015). After extracting object features, each object is concatenated to its tag. We generate a feature map for each image region, such that the entire visual features are encoded by a set of U image vectors as $F = \{f_1, f_2, \dots, f_U\}$, $f_i \in \mathbb{R}^d$. Therefore, we take a feature vector of dimension d for each image area. Before forwarding the visual features into the encoder, they are forwarded through a linear layer to adjust their dimension with the encoder.

A key issue in encoder-decoder architectures is appropriately assigning attention weights to relevant candidate vectors (Guo et al., 2020). Most existing Self-Attention (SA) models (Khan et al., 2021; Vaswani et al., 2017), utilize the Scaled Dot Product Attention (SDPA) (Vaswani et al., 2017; Wang et al., 2022) for the image captioning task. The SDPA is efficient and faster, and it can be conducted by the matrix multiplication techniques. The input contains three matrices, i.e., queries Q , values V , and keys K . The SA module computes the similarity between Q and K first, and then uses a softmax layer to produce the attention weights W as:

$$Attention(Q, K, V) = WV = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

where $Q \in \mathbb{R}^{L \times d_k}$ denotes the query, $K \in \mathbb{R}^{L \times d_k}$ is the key, $V \in \mathbb{R}^{L \times d_v}$ denotes the value, and the L represents the size of the sequence. $QK^T \in \mathbb{R}^{L \times L}$ denotes the product matrix, which could force the output to become too small or too large. This will affect the correctness of the parameters. So, d_k is utilized to scale QK^T .

In the encoder, N blocks are stacked in sequence to extract the improved features V^1 . Finally, a caption C is generated by the encoder-decoder structure as follows:

$$F = \text{FasterRCNN}(I) \quad (2)$$

$$V^1 = \text{Encoder}(F) \quad (3)$$

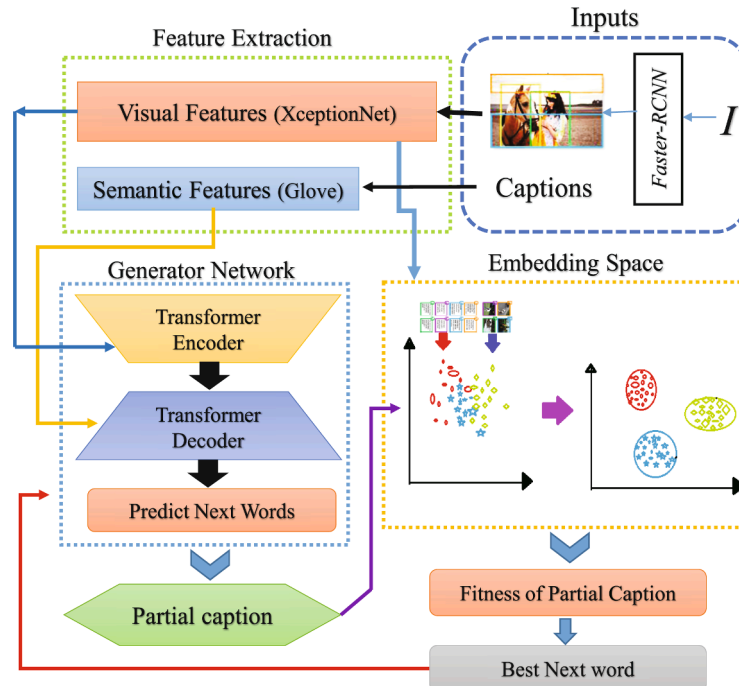


Fig. 3. The overall illustration of our framework.

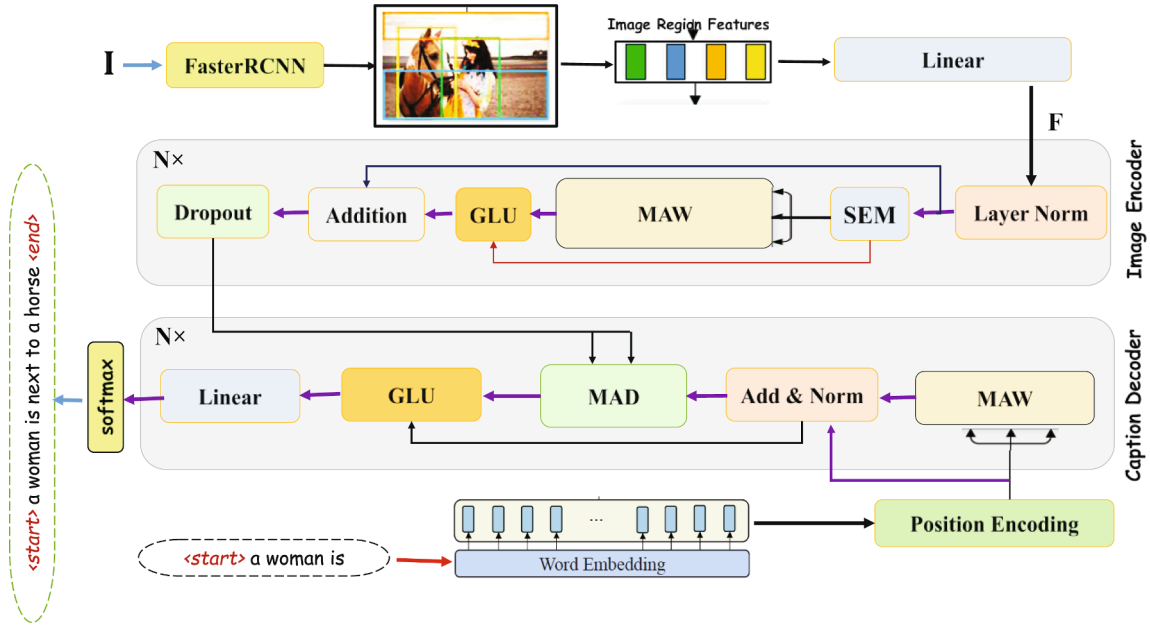


Fig. 4. Structure of the proposed generator network.

$$C = \text{Decoder} (V^l) \quad (4)$$

The proposed generator network improves the standard SA (Guo et al., 2020) by using different modules, including the Mask Attention Weight (MAW) module and Self-Elimator Module (SEG), in terms of caption generation. The MAW helps the SA to attend on the most appropriate information, while the SEM filters out some irrelevant information in the object features. Besides, we employ Gated Linear Units (GLU) (Dauphin, Fan, Auli, & Grangier, 2017) to tackle the concatenation of outputs of SEM and MAW modules for filtering unrelated details. Therefore, we concatenate the outputs of the MAW module and SEM, and then these outputs are forward to the GLU to omit the unrelated attention outputs. The encoding operation of the generator network can be described as follows:

$$V_{in} = \text{LayerNorm} (F) \quad (5)$$

$$V_{sem} = \text{SEM} (V_{in}) \quad (6)$$

$$Q, K, V = \text{Linears} (V_{sem}) \quad (7)$$

$$\hat{V} = \text{MAW} (Q, K, V) \quad (8)$$

$$V^l = \text{DropOut} (V_{in} + \text{GLU} ([V_{sem}, \hat{V}])) \quad (9)$$

where $\text{Linears}(\cdot)$ is three linear projections of V_{sem} , $[\cdot]$ denotes the concatenation operation. Note that, we utilize the dropout layer between the connections of each sub-encoder to relieve the over-fitting problem (Deorukhkar & Ket, 2021). Thus, the dropout layer makes our structure better generalize to the small image dataset. Additional details of each encoder module, such as LayerNorm, SEM, and MAW, are presented in their corresponding section. The decoding operation of the generator network can be formulated as follows:

$$\hat{V} = \text{MAW} (Q', K', V') \quad (10)$$

$$Z = \text{AddNorm} (\hat{V}) \quad (11)$$

$$V_{had} = \text{MAD} (W_q Z, W_k V^l, W_v V^l) \quad (12)$$

$$C = \text{Softmax} (\text{Linear} (\text{GLU} ([V_{had}, Z]))) \quad (13)$$

where $W_q, W_k, W_v \in \mathbb{R}^{d \times d}$ represent parameters to be learned. The proposed Multiple Attention Distribution (MAD) module provides multiple distributions to avoid a single shallow attention distribution and considers deeper internal relationships. Additional details of the MAD module, is presented in its corresponding section.

3.1.1. Encoder with pre-layer normalization

As shown in Fig. 4, the normalization layer over the input visual features is utilized first. Inspired by the recent development of transformer (Xiong et al., 2020), we introduce a pre-layer normalization transformer to seek a more efficient strategy to use the transformer for caption generation. Hence, for the pre-layer normalization transformer as represented in Fig. 5, we eliminate the feed-forward layer and utilize the norm-layer before transformer attention modules to simplify our structure for caption generation.

Compared with the structure of the original transformer in Fig. 2, the pre-layer normalization transformer is more effective and needs fewer

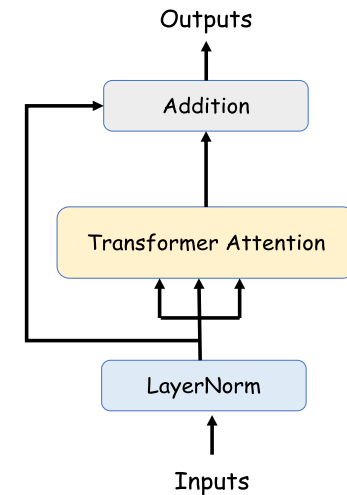


Fig. 5. Structure of the Pre-Layer Normalization. Compared with the structure of the original transformer in Fig. 2, the pre-layer normalization transformer is more effective and needs fewer parameters.

parameters. In addition, it reduces the necessary warm-up step for transformers in image description (Jiang, Li, Hu, Lu, & Liu, 2021; Pan, Yao, Li, & Mei, 2020) and won't reduce the performance. Hence, our pre-layer normalization simplifies the transformer structure for object feature enhancement. Technically, by stacking the attention layer and normalization layer, our model can learn relations among features effectively.

3.1.2. Encoder with SEM

Most recent image description approaches utilize the FasterRCNN to extract image region features. However, the detected regions by the bounding box include some unrelated information (e.g., areas of other objects or background). Besides, the extracted features are directly utilized by AM. Therefore, in our encoder, the Self-Eliminator Module (SEM) is used to eliminate some irrelevant details in the feature vectors.

The SEM considers the intra-object AM to remove the unrelated object features. For instance, the intra-object attention is generated for a feature vector set f , by projecting f through an embedding layer, followed by a sigmoid function Φ . Finally, SEM produces the feature vectors by stacking another linear projection of f with intra-object attention weight as:

$$SEM(F) = \Phi(W_1 f) \odot W_2 f \quad (14)$$

where W_1 and W_2 are learnable matrices. Note that the SEM does not change the dimension of f , and thus can easily be employed in the MAW module.

3.1.3. Encoder with MAW

The softmax layer in Equation (1) assigns the corresponding weights to all value vectors, even if the key and query vectors have no relevancy. To relieve this problem, we present the Mask Attention Weight (MAW) module to restrain the results of the softmax layer.

As illustrated in Fig. 6, by passing the value vectors and attention weights W through the MAW module, the attended vectors eliminate the trivial elements, while retaining the valuable components. Therefore, the MAW module directly constrains the attention weights because the exponential function of the softmax layer enlarges the distance among components.

The MAW module first calculates the weight value W_{ij} using the threshold operation by multiplying the mask R . Then, it normalizes every row of the final matrix as follows:

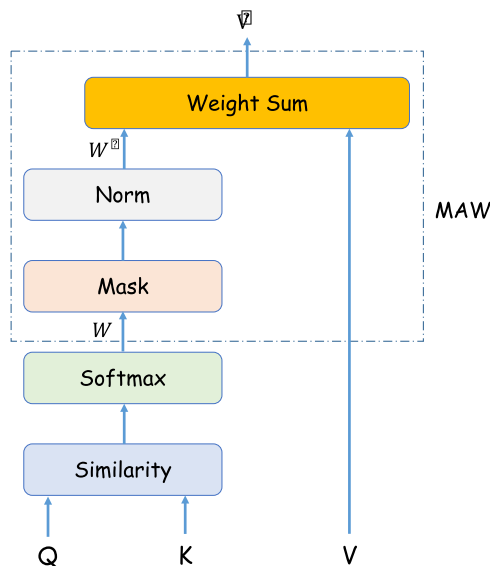


Fig. 6. The structure of the MAW module.

$$R_{ij} = \begin{cases} 0, & W_{ij} < e_i \\ 1, & W_{ij} \geq e_i \end{cases} \quad (15)$$

$$W'_{ij} = \text{Norm}(W_{ij} R_{ij}) = \frac{W_{ij} R_{ij}}{\sum_{j=1}^n W_{ij} R_{ij}} \quad (16)$$

where j, i denote the column and row of matrix respectively, and $E = (e_1, \dots, e_n)$, e_i denotes the i -th average value of the i -th row of W . The target output $\hat{V}$ is calculated as:

$$\hat{V} = V \odot W^r \quad (17)$$

where $\odot$ is element-wise operation, and W^r indicates the masked attention weights. Finally, the MAW module is formulated as:

$$MAW(W, V) = V \odot \text{Norm}(WR) \quad (18)$$

Note that, the MAW block can be seamlessly employed into SA without raising any parameters.

3.1.4. Decoder with MAD

The transformers calculate the correlation between the q and k vectors. These vectors are classified into several sub-vectors to compute their correlation, while all are separate. Hence, the distribution of the weights for the sub-vectors is independent (Ye et al., 2018). In addition, SA includes only a single distribution for predicting the attention weights, and this limits the performance of the method. As mentioned above, the basic transformer-based AM is as follows:

$$\text{Attention}(q_j, k_j, v_j) = f_{att}(q_j, k_j) v_j \quad (19)$$

where f_{att} calculates the correlation between two vectors. Equation (19) is a desirable measure for calculating the similarity between vectors, while it is complicated to measure the distance between vectors in various modalities.

To relieve these issues, we present a Multiple Attention Distribution (MAD) module for utilizing the weight distribution in the SA, and measuring the distribution of the weights between the k and q vectors in the sub-space. As illustrated in Fig. 7, for each sub-space, we utilize a weighted summation of their attention distribution, to acquire MAD. Moreover, the MAD is performed as parallel to make the model relate to every sub-space by substituting the single distribution.

In the decoder part, we first divide the high-dimensional feature vector into multiple low-dimensional sub-spaces. Then, we connect these independent sub-spaces to boost the model's capability.

The function f_{att} in Eq. (19) superimposes multiple intrinsic inter-vector similarities to represent complex relations, and needs several additional parameters to enable the module to process complex multi-modal data. As described in Fig. 7, the colored boxes demonstrate the attention distribution for every sub-space. Therefore, the final output for f_{att} is obtained utilizing the weight matrix Ω as:

$$f_{att}(q_j, k_j) = \text{softmax}\left(\sum_{i=1}^h \Omega_{ji} u_i\right), \quad j = 1, 2, \dots, h \quad (20)$$

in which h denotes the set of heads and $\Omega \in \mathbb{R}^{h \times h}$ represents model parameters which can calculate the similarity between heads. In addition, $U = \{u_1, u_2, \dots, u_h\}$ calculates the correlation between vectors in multiple heads as:

$$u_j = \frac{q_j k_j}{\sqrt{d_k}}, \quad j = 1, 2, \dots, h \quad (21)$$

where d_k represents the sub-space dimension. Before measuring u_j , we normalize k_j and q_j to provide numerical stability. Thus, we determine the expression of each head $j \in \mathbb{R}^{L \times d_k}$ as:

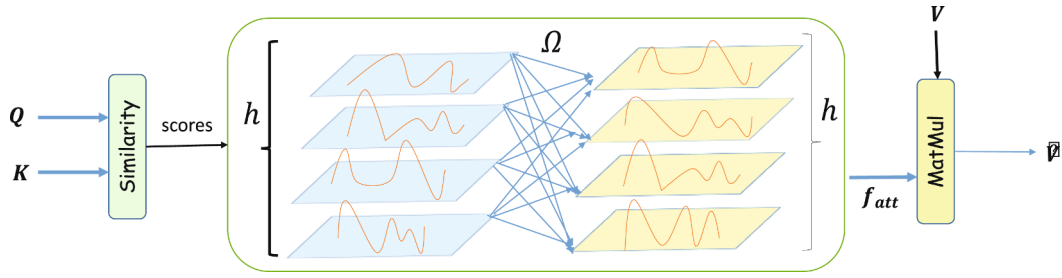


Fig. 7. The visualization of the MAD module.

$$head_j = \text{Attention}(q_j, k_j, v_j), \quad j = 1, 2, \dots, h \quad (22)$$

At last, we obtain the final expression for the MAD as:

$$\text{HAD}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(head_1, head_2, \dots, head_h) \quad (23)$$

where the function *Concat* is the concatenate function. As a result, the MAD module takes the queries from the previous sub-layer, and the values and keys from the output of the encoder. This enables the decoder to attend on all regions in the input sequence.

3.1.5. Training generator network

Similar to the training process of most image description methods (Zhang et al., 2021; Zhu, Li, Liu, Li, et al., 2018; Zhu, Li, Liu, Peng, et al., 2018), we train the generator network utilizing two optimization algorithms. We employ the cross-entropy loss (XE) to pre-train the proposed method and utilize Reinforcement Learning (RL) to fine-tune sequence generation. First, the generator network is trained utilizing the XE as:

$$L_{XE} = - \sum_{t=1}^T \log(p_\theta(w_t | w_{1:t-1})) \quad (24)$$

where $w_{1:t-1}$ is the target GT sequence, and the T denotes the length of the sequence. The probability p_θ is defined by parameter θ of the model. Next, we optimize the model by CIDEr reward utilizing the Self-Critical Sequence Training (SCST) (Rennie, Marcheret, Mroueh, Ross, & Goel, 2017) as:

$$L_{RL}(\theta) = -E_{w_{1:T} \sim p_\theta}[r(w_{1:T})] \quad (25)$$

where $r(\cdot)$ indicates the CIDEr reward. The gradients are approximated as:

$$\nabla_\theta L_{RL}(\theta) \approx - (r(w_{1:T}) - r(\hat{w}_{1:T})) \nabla_\theta \log p_\theta(w_{1:T}) \quad (26)$$

where the $w_{1:T}$ represents a sampled sentence, and $r(\hat{w}_{1:T})$ is the baseline reward provided by greedily decoding the current model. The SCST investigates the space of sentence by sampling from the policy p_θ during

training. These gradients tend to enhance the probability of sampled sentences that reward higher than the reward from the current model.

3.2. Multi-modal selector module

In this section, a selector network as the exploitation of the fitness for the Partial Caption (PC) is presented. The proposed selector module takes the top-ranked words from the generator network outputs at each state and predicts the next best word.

As represented in Fig. 8, the proposed selector module has three key parts. In the first step, we use the semantic and visual inputs to create the embedding space. Then, in the second step, we train a latent space to project captions and images into the same embedding space to learn the sentence-image similarity. Finally, in the third step, we train a Multi-Linear Perceptron (MLP) to evaluate the PC by assigning fitness scores through the embedding space. Additional details for each step are presented in the following.

In the first step, to create the embedding space, we utilize the Xception network (Chollet, 2017) for feature extraction from images. Moreover, the Glove (Pennington, Socher, & Manning, 2014) is utilized for encoding the semantic content of a partially extracted caption.

Given an image I_i , let X_i^- and X_i^+ show its sets of non-relevant and relevant descriptions, respectively. For a non-relevant caption x_k and a relevant caption x_j , we need the distance between I_i and x_j to be smaller than the distance between I_i and x_k , with a radius of u by utilizing the triplet-wise constraint as:

$$D(I_i, x_j) + u < D(I_i, x_k) \forall x \in X_i^+, \forall x_k \in X_i^- \quad (27)$$

Besides, for an input caption x_i , we have similar equations in the other direction as:

$$D(I_j', x_i) + u < D(I_k', x_i) \forall I_j' \in I_i^+ \forall I_k' \in I_i^- \quad (28)$$

where I_i^- and I_i^+ are the non-relevant and relevant image features for x_i . In the second step, we train the embedding space using all the triplets

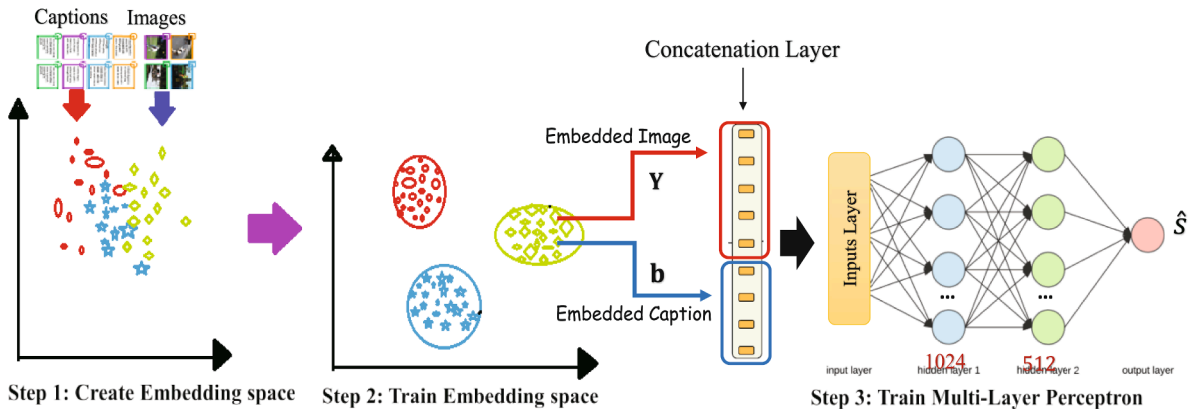


Fig. 8. The structure of the proposed selector module.

formulated in Eq. (27) and Eq. (28) as follows:

$$L(I, X) = \lambda \sum_{i,j,k} [u + D(I_i, x_j) - D(I_i, x_k)]_+ + (1 - \lambda) \sum_{i',j',k'} [u + D(I_{i'}, x_{j'}) - D(I_{i'}, x_{k'})]_+ \quad (29)$$

where $[Q]_+ = \max(0; Q)$ and the weight λ determine the effect of the loss values in each direction.

We set the radius $u = 0.3$ for total terms in our experimental evaluations. Optimizing Eq. (29) requires enumerating triplets that it is mathematically expensive for a big database. Inspired by (Liu, Zhu, Ye, Guadarrama, & Murphy, 2017; Ren, Wang, Zhang, Lv, & Li, 2017), we utilize a gradient descent algorithm to optimize the sample triplets in each mini-batch and loss function.

As illustrated in Fig. 8, after training the embedding space for an embedded caption $\mathbf{b}$ and an embedded image $\mathbf{Y}$, we calculate a fitness score to be the embedding similarity between $\mathbf{Y}$ and $\mathbf{b}$ as:

$$s = \frac{\Phi(\mathbf{Y}) \cdot \Psi(\mathbf{b})}{\|\Phi(\mathbf{Y})\| \cdot \|\Psi(\mathbf{b})\|} \quad (30)$$

in which s is the fitness score of the generated partial caption. Φ and Ψ are the mapping functions from image/caption features to the embedding space.

Finally, in the third step, a three-layer MLP with 1024 and 512-dimensional hidden layers is trained that produce a fitness score $\hat{s}$. Therefore, the proposed selector network is trained by minimizing the Mean Squared Error (MSE) as:

$$MSE = \|\hat{s} - s^2\| \quad (31)$$

As a result, the selector module, which calculates the distance between sentences and images, can determine the correctness of generated sentences and performs global guidance for caption generation.

3.3. Image captioning

In the proposed framework, caption generation is performed by sentence-level and word-level strategies. Hence, the generator network and the selector module are utilized to extract the best word based on multi-modal information. Similar to the beam search mechanism (Farhadi et al., 2010; Yang et al., 2011), the selector network takes the top-ranked words (Top K) of the generator network outputs and utilizes them to select the next best word at each state; where K denotes a hyperparameter and denotes the number of words with the highest score. Therefore, the selector module employs all possible next words and predicts the best word at each state.

The generator network determines the word confidence, which locally produces next words as top K . Conversely, the selector module seeks caption confidence, which globally analyzes the current state to select the next best word. These complement confidences are collaboratively employed to regulate the distribution of the next word which is similar to GT.

3.4. Algorithm design

The pseudocode of the presented model is given in Algorithm 1. There are several matrices and parameters as inputs in our algorithm. These inputs are images I , captions C and model hyperparameters. The training process of our algorithm requires that we choose a loss function and an optimization algorithm.

The proposed framework has three key steps. In the first step, after configuring the hyperparameters of the algorithm (line 1), the generator module is trained (lines 2–7). Then, in the second step, we train the embedding space by utilizing the appropriate objective function (lines 8 and 9).

In the third step, our selector module is trained by looping across the

data iterator, and forwarding the data to the model. In each iteration, we get the outputs of the embedding space and the generator network (lines 10–12) to train MLP using the MSE loss (lines 13 and 14) and back-propagate the loss through the network (line 15). Finally, we update the network to decrease loss values (line 16). As a result, the training step continues until all images are given to the algorithm (lines 10–16). Finally, for caption generation, the learned networks are utilized to predict the best word at each state.

Algorithm 1. Pseudocode of the proposed framework

Inputs: I, C
Outputs: Output captions
begin:

- 1: **Initialize** $K, \text{num_epochs}$
- Step 1: Training the GeneratorNetwork**
- 2: $F = \text{Faster-RCNN}(I)$
- 3: $V^l = \text{Image-Encoder}(F)$
- 4: $C_{out} = \text{Caption-Decoder}(C, V^l)$
- 5: $\text{Loss} = \text{Criterion}()$ using Eq. (24)
- 6: $\text{Loss.Backward}()$
- 7: **Optimizer.** $\text{Step}()$
- Step 2: Training the EmbeddingSpace**
- 8: **Create** an Embedding space
- 9: **Train** the Embedding space using Eq. (29)
- Step 3: Training the SelectorModule**
- 10: $\text{TopK_words} = \text{GeneratorNetwork}(F, C)$
- 11: $B = \text{Create a set of partial captions from TopK_words}$
- 12: $\text{Embedded_vectors} = [\text{EmbeddingSpace}(b, I) \text{ for } b \text{ in } B]$
- 13: $\hat{s} = \text{MLP}(\text{Embedded_vectors})$
- 14: $\text{Loss} = \text{Criterion}(\hat{s}, s)$ using Eq. (31)
- 15: $\text{Loss.Backward}()$
- 16: **Optimizer.** $\text{Step}()$

End

4. Experimental results

In this section, we carry out various experiments to measure the performance of our framework. Firstly, we discuss the experimental settings and datasets. Then, we evaluate our framework compared with state-of-the-art frameworks for the image description task. Finally, we perform various analyses of the proposed model.

4.1. Experimental settings

4.1.1. Datasets

We analyze our model¹ on Flickr30k² (Bai & An, 2018), and MSCOCO³ (Chen et al., 2015) datasets. The Flickr30k dataset contains 31,783 images, while the MSCOCO dataset contains 123,287 images. Each image is captioned with five sentences. The MSCOCO dataset is more challenging because the images consist of various objects in their natural context.

Since there is no methodized split on these datasets, we perform the popularly split (N. Yu, Hu, Song, Yang, & Zhang, 2019; J. Zhang, Mei, Zheng, & Fan, 2021; L. Zhou et al., 2020) on both Flickr30k and MSCOCO datasets for a fair comparison. For instance, 5000 images are used for testing, 5000 for validation, and the rest for training in MSCOCO. Besides, 1000 images are used for testing, 1000 for validation and the rest for training in Flickr30k.

4.1.2. Evaluation measures

We evaluate the models utilizing several well-known measures, including BLEU (Papineni, Roukos, Ward, & Zhu, 2002), METEOR (Denkowski & Lavie, 2014), ROUGEL (Lin, 2004) and CIDEr (Vedantam, Lawrence Zitnick, & Parikh, 2015). BLEU uses the n -gram precision and METEOR utilizes the harmonic mean of unigram precision and recall.

¹ <https://github.com/h-parvin/ImgCaptioning.git>.

² <https://www.kaggle.com/datasets/hsankesara/flickr-image-dataset>.

³ <https://cocodataset.org/#download>.

Unlike METEOR, BLEU employs relationships at the corpus level. ROUGE-L is utilized to seek the usual sub-sequence with the highest length between the source and target captions. CIDER is employed to assess image captioning utilizing human consensus.

Higher values are given better accuracy for all these measures. All these measures calculate the consistency of n -grams between reference captions and extracted captions. For fair comparisons with the competent models, we employ the openly usable evaluation code introduced by MSCOCO Server (Chen et al., 2015) to evaluate our model.

4.1.3. Data pre-processing

In our implementations, we utilize the standard pre-processing method for the captions and images (Vinyals, Toshev, Bengio, & Erhan, 2017). For instance, we change each caption to lowercase and remove the total non-alphabetic characters. We discard words that occur less than three times in Flickr30k or five times in MSCOCO. Thus, we obtain a vocabulary with sizes 7652 and 10,478 in Flickr30k and MSCOCO, respectively.

Furthermore, we attach an end-token <End> and a start-token <Start> to the vocabulary. During the evaluation, we fix the maximum caption length as 30. For image features, we extract them utilizing the outputs of the last layer of the Xception network. In addition, we utilize FasterRCNN to represent image regions. The feature vector for each region is 2048-dimensional. Besides, we utilize Glove (Pennington et al., 2014) as a word embedding model and linearly project to model input dimensional to represent words. We set the word embedding size to 512.

4.1.4. Implementation details

In the implementation, we employ the beam size method to enhance the accuracy of the output caption with sizes 2 for Flickr30k and 5 for MSCOCO. Adam (Devi, Thiruvikraman, Kashyap, & Shylaja, 2020) is utilized to optimize our algorithm with the learning rate as 0.0001 and decaying every-two epochs by a factor of 0.6. We tune batch sizes 100 and 50 to both generator and selector networks, respectively. To reduce over-fitting, we utilize dropout with a rate of 0.4 and early-stopping if the validation CIDEr rate increases over the last 15 iterations. We fix the maximum epoch number as 100000. In addition, we only utilize the captions and images in the dataset to train the proposed model, without any external data.

We follow the transformer-based methods such as (Li et al., 2019; Liu et al., 2021; Zhang et al., 2021) to adjust the hyperparameters for a fair comparison. Moreover, the dimensionality of target image vectors is 2048. The latent dimension in the MAW module is 512, and the set of heads is 8. We utilize the same number of layers 6 for the decoder and encoder. In the training process, we employ a step decay schedule for updating the learning-rate. Moreover, we adjust the model size which is d_{model} to be 512 and the mini-batch to be 16. The initial learning-rate of the transformer is 0.0004 and the weight-decay is 0.6. All experiments are performed on NVIDIA GPU Tesla T4 utilizing python programming language.

4.2. Comparisons with state-of-the-art models

To further validate the accuracy of our approach, it is compared with several competent and state-of-the-art models including VIS SAS (Zhou et al., 2020), NEURALTALK (Yu et al., 2019), INJECT-TAG (Zhang et al., 2021), BASED-GENERATION (Ding et al., 2019), DSEN (Xiao et al., 2019), AVTMN (Song, Zhu, & Jiang, 2020), SEM (Cai & Liu, 2020), ENSCAPTION (Yang et al., 2020), LWDSFUSION (Tang, Wang, & Kwong, 2018), SCST (Zhou, Wang, Liu, Hu, & Zhang, 2020), SEN (He, Tavakoli, Borji, & Pugeault, 2019), PAGNET (Song, Liu, Qian, & Chen, 2019), GAT (C. Wang et al., 2022), ALT (Ye et al., 2018), BT (J. Li et al., 2019) and TRANSKG (Zhang et al., 2021). Since the outputs of these models for image description were presented publicly in the literature, the proposed framework is compared with these state-of-the-art models

directly.

In Tables 1 and 2, we show the results of recent state-of-the-art frameworks on the benchmark datasets, including COCO and Flickr datasets under seven different metrics. In the tables, the best results are demonstrated in bold-faced style. As the Tables show, our framework outperforms the others in most cases. For instance, compared with attention-based models, such as ENSCAPTION, SEN and PAGNET, our model obtains better performance across most metrics. Therefore, unlike previous attention learning without the transformer, our model yields a noticeable improvement by searching the effective image captions with a reasonable guiding strategy.

Besides, the state-of-the-art transformer-based models, including TRANSKG, ALT, BT and GAT models achieve higher accuracy than the NIC methods across most metrics. As a result, the transformer module can get higher accuracy relative to the LSTM-based models. Hence, the transformer module can be employed in the image description task successfully, and has better performance than recurrent-based models such as LWDSFUSION, VIS SAS, NEURALTALK, BASED-GENERATION and AVTMN. Still, compared with transformer-based models including TRANSKG, ALT, BT and GAT models, the proposed model could achieve a higher score across most metrics.

BLEU calculates the similarity between a set of reference captions and the generated caption to validate the quality of our generated sentence. The BLEU scores in Table 1 denote how similar the candidate caption is to the reference captions. For example, the proposed model achieved 5.9 %, 7.8 %, 6.9 %, and 3.7 % average improvement in BLEU 1, BLEU 2, BLEU 3 and BLEU 4, respectively, on the Flickr dataset.

METEOR is based on word to word matches through the use of corresponding word synonyms and stems. Unlike the BLEU which seeks correlation at the corpus-level, METEOR generates a good correlation with human judgment at the sentence-level. For instance, the proposed model improves the METEOR by about 4.6 % and 11.2 % over the Flickr30k and MSCOCO datasets, respectively.

CIDEr weights each n -gram in a caption based on its frequency to calculate similarity between reference captions and predicted captions. Hence, CIDEr takes human judgment of consensus better than existing measures across captions produced by various sources. From the Tables, we can find that the proposed method achieved 15.3 %, and 22.5 % average improvement in CIDEr, over the Flickr30k and MSCOCO datasets, respectively.

To measure sentence similarity, ROUGE-L utilizes sentence-level structure similarity and determines longest co-occurring in sequence n -grams naturally. For instance, the proposed model improves the ROUGE-L by about 7.3 % and 12 % over the Flickr and COCO datasets, respectively.

The comparison of the results demonstrates that employing transformer-based structures leads to performance improvement in image captioning. Therefore, we can conclude that our framework has state-of-the-art results due to the use of MAW, SEM MAD and selector modules in terms of caption generation. The MAW module helps the SA to attend on the most appropriate information. Furthermore, the object detection features help the decoder determine which regions of the input image are relevant during image description. Besides, the SEM filters out some irrelevant information in the object features. Moreover, the MAD module provides multiple distributions to avoid a single shallow attention distribution and considers deeper internal relationships. In addition, the selector module determines the correctness of generated sentences and performs global guidance for caption generation.

Therefore, by exploiting word-level (local) and sentence-level (global) policies, the proposed method achieves a higher score across most metrics. All in all, these comparisons illustrate the superiority of the proposed algorithm.

4.3. Sensitivity analysis

Similar to the beam search mechanism, our selector network gets the

Table 1

The comparisons on the Flickr30k dataset with beam size 2. (–) denotes that the value is not reported. The results are provided as percentage (%), with the highest value in bold-face.

Models	BLEU 1	BLEU 2	BLEU 3	BLEU 4	METEOR	ROUGE-L	CIDEr
VIS_SAS	66.4	44.0	30.1	20.7	18.4	47.9	38.9
NEURALTALK	64.6	43.8	31.9	22.4	19.2	43.8	39.6
INJECT-TAG	68.7	49.3	35.0	24.9	24.7	53.4	45.0
BASED-GENERATION	67.3	47.8	32.6	22.1	19.5	45.9	44.4
DSEN	–	–	39.5	28.7	21.9	–	68.7
AVTMN	–	–	–	24.8	20.8	49.4	59.8
SEM	73.1	55.1	40.1	29.0	22.0	50.1	66.8
ENSCAPTION	76.2	58.7	43.5	32.9	24.6	54.2	69.3
LWDSFUSION	65.1	47.2	33.4	23.8	20.5	47.0	50.8
SCST	73.4	–	–	30.1	22.6	–	69.3
SEN	–	–	–	22.0	18.4	44.1	41.6
PAGNET	74.8	55.8	41.2	30.7	25.2	61.4	57.5
GAT	74.4	56.7	41.8	30.8	23.4	–	68.0
TRANSGK	68.3	–	–	26.5	21.7	47.9	56.6
ALT	65.5	50.7	37.0	27.0	21.2	48.0	56.2
Proposed method	75.8	58.8	43.9	30.2	26.3	56.8	70.8
Avg. Improvement	5.9 %	7.8 %	6.9 %	3.7 %	4.6 %	7.3 %	15.3 %

Table 2

The comparisons on the MSCOCO dataset with beam size 5. (–) denotes that the value is not reported. The results are provided as percentage (%), with the highest value in bold-face.

Models	BLEU 1	BLEU 2	BLEU 3	BLEU 4	METEOR	ROUGE-L	CIDEr
VIS_SAS	72.4	52.5	38.1	28.0	23.6	55.4	82.1
NEURALTALK	74.0	56.7	43.3	31.3	25.5	53.4	98.3
INJECT-TAG	76.1	59.1	44.4	33.5	30.1	60.5	95.1
BASED-GENERATION	76.8	60.5	45.8	34.2	26.1	55.5	105.5
DSEN	75.3	59.6	45.7	34.7	26.2	55.6	106.8
AVTMN	–	–	–	33.1	27.3	56.7	112.6
SEM	80.0	64.3	49.6	37.5	28.2	58.2	126.0
APTION	81.7	65.3	51.1	39.2	29.4	59.0	125.5
LWDSFUSION	73.1	56.5	42.2	31.3	25.7	54.0	99.9
SCST	80.2	–	–	38.0	28.5	–	126.1
SEN	–	–	–	30.3	23.8	51.8	90.7
PAGNET	83.2	62.8	46.3	40.8	30.4	58.6	118.6
GAT	81.1	66.1	51.8	39.9	38.4	59.1	129.8
TRANSGK	76.2	–	–	34.4	27.7	56.3	112.6
ALT	74.2	57.7	44.3	34.1	27.0	55.2	105.3
BT	80.5	65.2	50.6	39.7	29.4	59.1	131.5
Proposed method	86.1	66.5	49.1	37.8	39.2	65.1	132.9
Avg. Improvement	8.6 %	5.9 %	3.0 %	2.8 %	11.2 %	12.07 %	22.5 %

top-ranked words as K from the generator network outputs and utilizes them to select the best word at each state; where K is a hyper-parameter and denotes the number of words with the highest score.

Therefore, to verify the effect of beam size K for image description, in Figs. 9 and 10, we indicate the effect of various K values on both COCO and Flickr datasets. As mentioned in previous works (Vaswani et al., 2017; Vinyals et al., 2017; Wang et al., 2020), if K is large, the measure scores reduce the corresponding efficiency. Hence, we analyze this discovery for our model. We analyze our algorithm with K from 1 to 10. As a result, we can see that the effect of K is partially significant. That is, as we increase the K , inappropriate candidate words are more probable to be drawn into the beam.

As the Figures show, the proposed model is less sensitive to K and so the result variations for various K are relatively small. In other words, our model employs different transformer-based modules which are more robust and stable to change of K .

To make a tradeoff between accuracy and network complexity, we set the K , where the proposed model obtains its best performance. As a result, the value of K is tuned to its optimal values as $K = 5$ and $K = 2$ in MSCOCO and Flickr30k datasets, respectively.

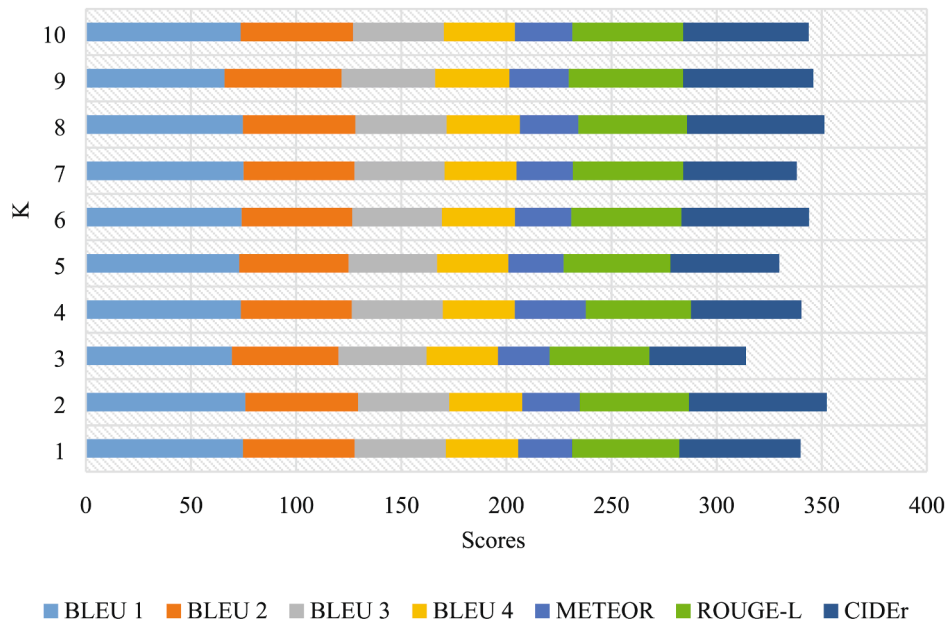
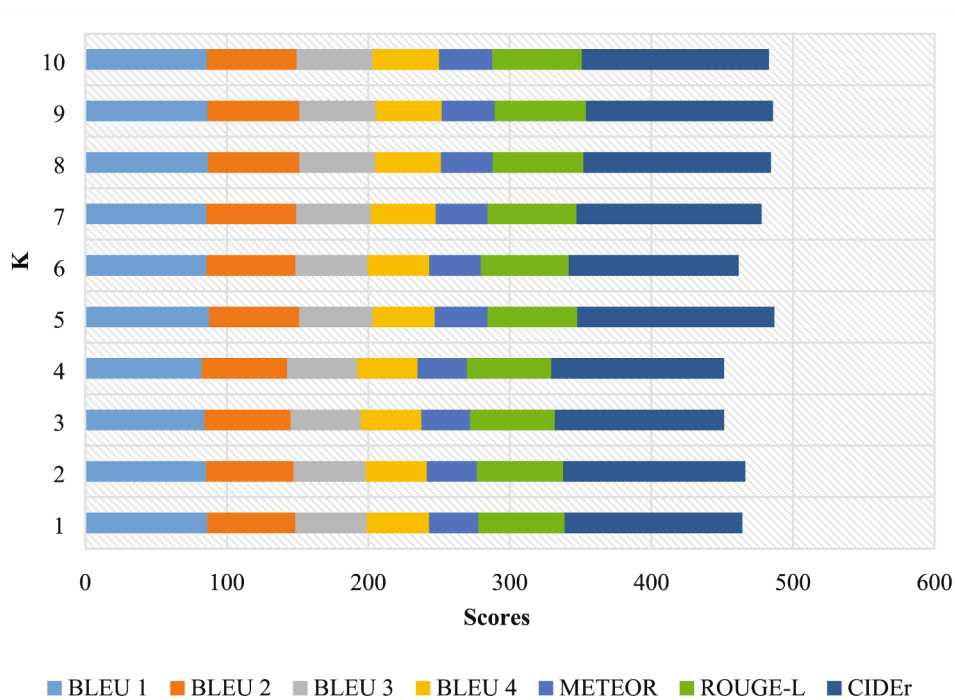
In addition, to verify the effect of λ in Eq. (29) for image description, we analyze the evaluation of λ impact on the proposed model. As in Eq. (10), λ is a hyperparameter determining the effect of the loss values in each direction, $0 \leq \lambda \leq 1$. As shown in Fig. 11, as λ goes down from 0.4

to 0 or goes up from 0.4 to 1, overall the performance drops. This determines the importance of the non-relevant and relevant image features. From the results, it is clear that the best accuracy is when $\lambda = 0.5$ and $\lambda = 0.4$ in MSCOCO and Flickr30k datasets, respectively.

Besides, we analyze the impact of number of encoder/decoder layers N , and kept the structure of encoder/decoder fixed as Fig. 4. From the results in Figs. 12 and 13, we see that increasing the number of layers enhances the performance of caption generation as the transformer learns a better representation of the word as it gets deeper. Therefore, increasing the number of layers improves the quality as the transformer gets to attend to information from different representation subspaces. However, increasing the number of layers may reduce the accuracy of the results, if the number of layers is too large, the complexity will increase significantly. To make a tradeoff between accuracy and network complexity, we set the N , where the proposed model acquires its best performance. As a result, the value of N is tuned to its optimal values as $N = 7$ and $N = 6$ in MSCOCO and Flickr30k datasets, respectively.

4.4. Effect of the selector module

In this section, we analyze the impact of the presented selector network for image description. Figs. 14 and 15 indicate the effect of the selector network across both datasets. From the results, we found that the impact on various measures for model performance is not the same

Fig. 9. Effect of K over the Flickr30K dataset.Fig. 10. Effect of K over the MSCOCO dataset.

and the effect of the selector module is very evident.

In other words, all evaluation measures are high when the selector network is employed to evaluate the partial captions. Thus, the experimental evaluations over well-known metrics indicate that employing the sentence-level policy can lead to accuracy enhancements.

The selector module learns a joint representation of description and image through the embedding space for generating fitness scores to select the next best word at each state. Thus, the sentence-level guidance is utilized in our model to help the transformer structure generate the related words which boosts the encoder-decoder structure.

4.5. Convergence behavior

This section evaluates the convergence behavior for both generator and selector networks. Hence, the number of epochs/steps needed for our model to converge is analyzed. Figs. 16–19 indicate the convergence curves of our model for the MSCOCO and Flickr datasets. As the Figures demonstrate, our model has a suitable convergence speed. For instance, Figs. 17 and 19 indicate that the proposed selector network has an appropriate convergence speed while it has a low loss value in Flickr and MSCOCO datasets.

Furthermore, we can see from the convergence curves that the

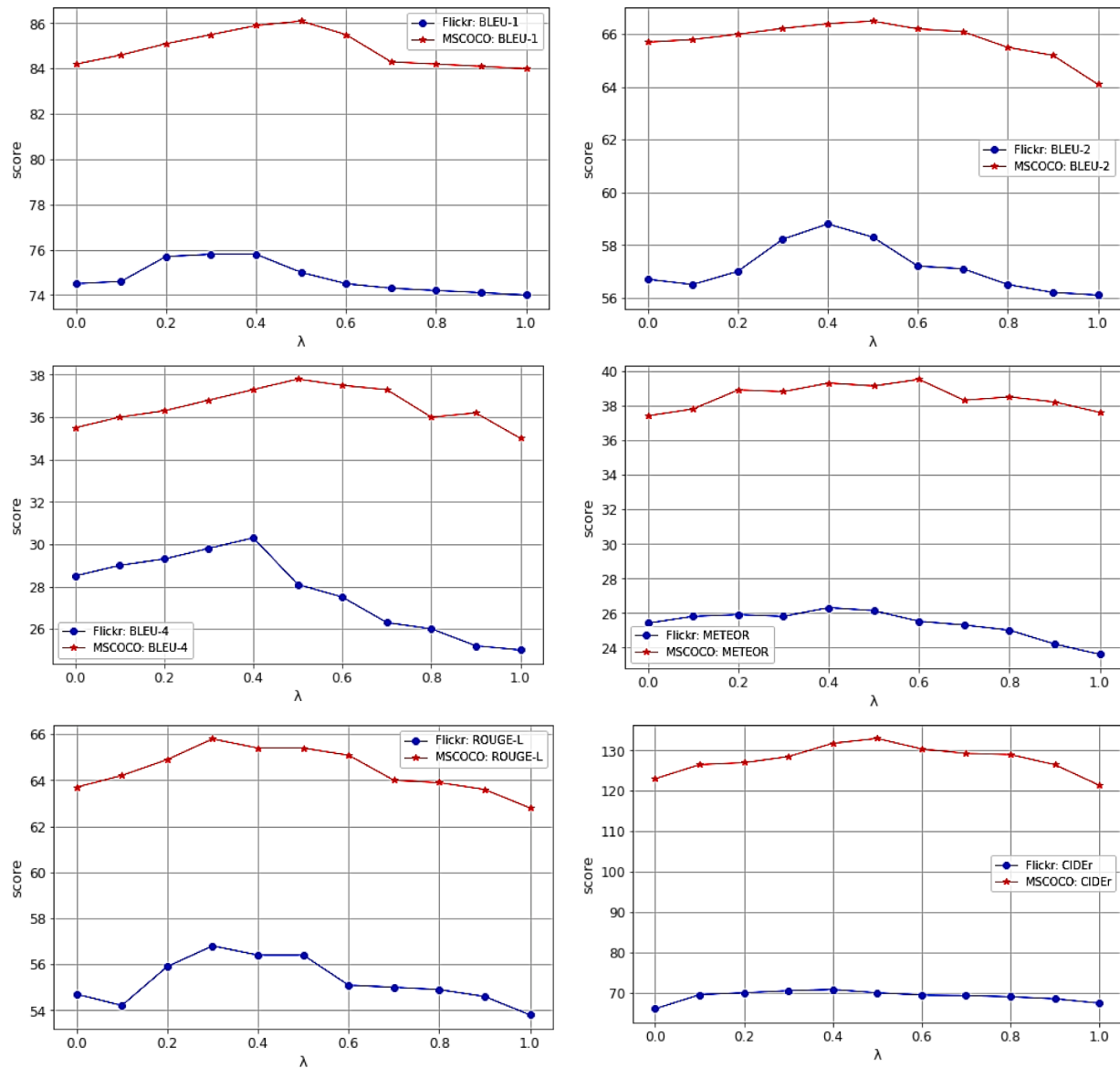


Fig. 11. Sensitivity of λ over the Flickr and MSCOCO datasets based on different evaluation measures.

proposed model could escape from the local optima in later iterations. For instance, the Figures show that our selector network generally needs only 120 epochs to converge in both MSCOCO and Flickr datasets. Moreover, the results in Fig. 16 show that convergence curves of the generator network become stable on the Flickr dataset after 40,000 steps (when the number of batch size is set to 50). In addition, the generator network slowly converges in Fig. 18 on the COCO dataset after 80,000 steps (when the number of batch size is set to 100).

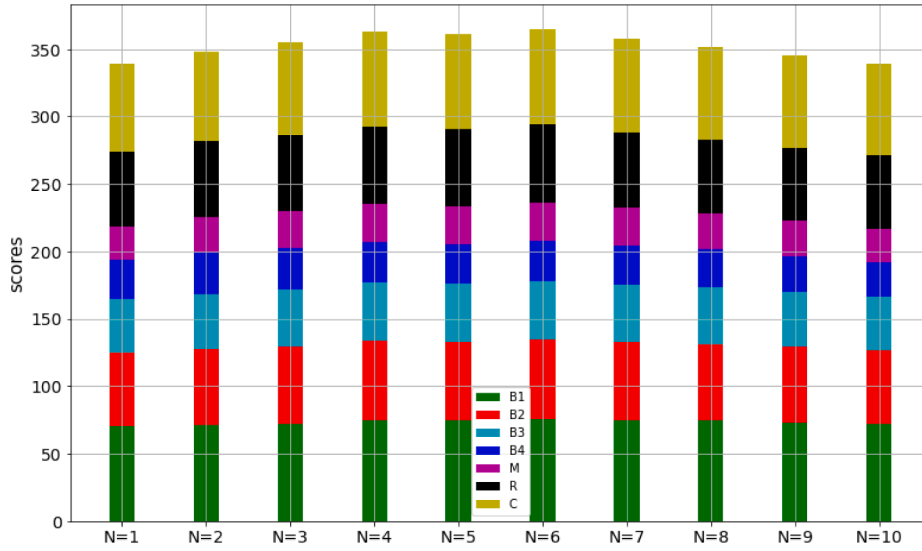
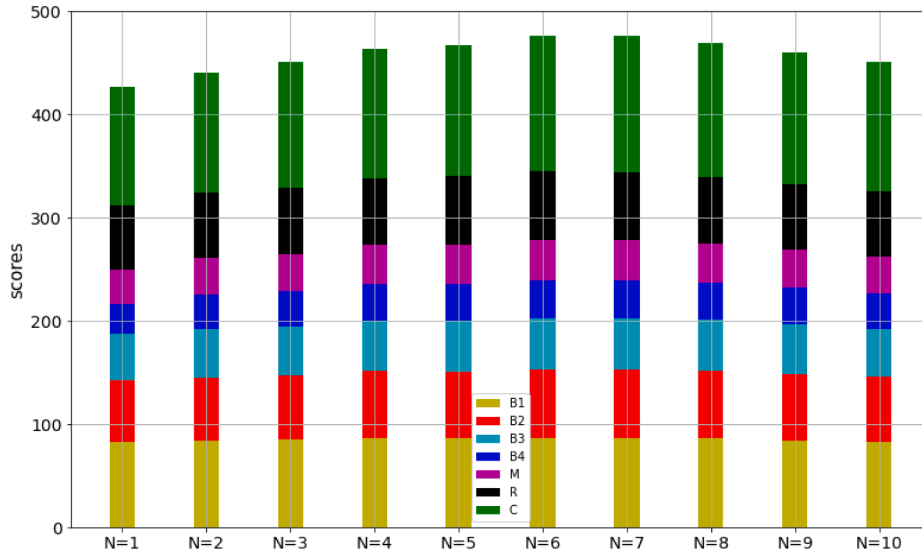
The proposed model utilizes the appropriate transformer-based structure in its process, which leads to a decrease in the number of epochs/steps required for obtaining stable results. In other words, we eliminate the feed-forward layer and utilize the norm layer before transformer attention modules to simplify the network. Therefore, the pre-layer normalization transformer is more effective and needs fewer parameters. As a result, this evidence indicates that our algorithm has appropriate convergence behavior, and it can be utilized effectively in practical applications.

4.6. Statistical significant test

In this section, a statistical *student's test* is performed to demonstrate

the strength of reported results to show if our model has significant differences from the other frameworks. The statistical test is employed to compare the differences between two sample means. For each dataset and each model, the results of the *student's test* are provided in Table 3. Therefore, the null hypothesis is determined as the enhancement of our method compared to other models is insignificant. If the *pvalue* is lower than a significant-level (here 0.05), the null hypothesis is rejected, which means our model reached significant accuracy compared to other models.

Besides, an alternative hypothesis is determined as the substantial difference in our model compared to a competent model. If the *pvalue* is lower than a significant-level, the hypothesis is accepted. The analysis demonstrates that in all cases except two (denoted as *), the *pvalues* of the other algorithms are lower than 0.05 which shows the alternative hypothesis is accepted and the null hypothesis is rejected. Hence, we inferred that our framework obtains statistically significant results compared to other models. As a result, we can conclude that the statistical experiments illustrate the considerable enhancement of the results achieved by our framework in comparison with other frameworks.

Fig. 12. Sensitivity of N over the Flickr30k dataset.Fig. 13. Sensitivity of N over the MSCOCO dataset.

4.7. Scalability analysis

Scalability refers to the capability of a model to work on large-scale dataset. Specifically, scalability means that as the size of data become larger, its performance enhances correspondingly. Due to the rapid growth of online information, the scalability of a model should be correctly studied. To this end, several experiments are conducted on Flickr and COCO datasets to prove the scalability of our model on large-scale data. As shown in Fig. 20, our model is employed on varying parts of each dataset from 0.1 to 1 percentages stepping by 0.1. We also show the average running time per epoch for varying parts of each training data. The experiments indicate that the training time enhances linearly by growing the size of the training data. This means that the proposed method can be employed to large-scale data. Note that the proposed method is trained on two NVIDIA GPU Tesla T4 utilizing data parallelism. Consequently, the comparison results prove the efficiency of the proposed model in enhancing model performance and decreasing computational overhead.

4.8. Qualitative results

In this section, we illustrate some concrete captions predicted by the proposed model. Fig. 21 shows the tested images and the corresponding generated captions over both datasets. For each sample, there are two different captions. Concretely, the “G” indicates GT captions, and “P” shows generated captions by our model. As demonstrated by this example, our model efficiently attends to fine details or large image areas. This ability comes because the AM in the proposed approach contains various important areas with related objects. Moreover, our model can employ all of the information about an object.

Besides, image descriptions given by our method have a high relationship with visual information of the corresponding image that varies from human activities to other things. Furthermore, the newly extracted descriptions have new objects or phrases that are not defined in the GT captions, while matching the visual data of images. Overall, we can see that our framework can correctly identify objects. At last, we can conclude that the proposed model can make the system produce descriptions using local-global guidance, in which the generated sentences demonstrate details of a query image.

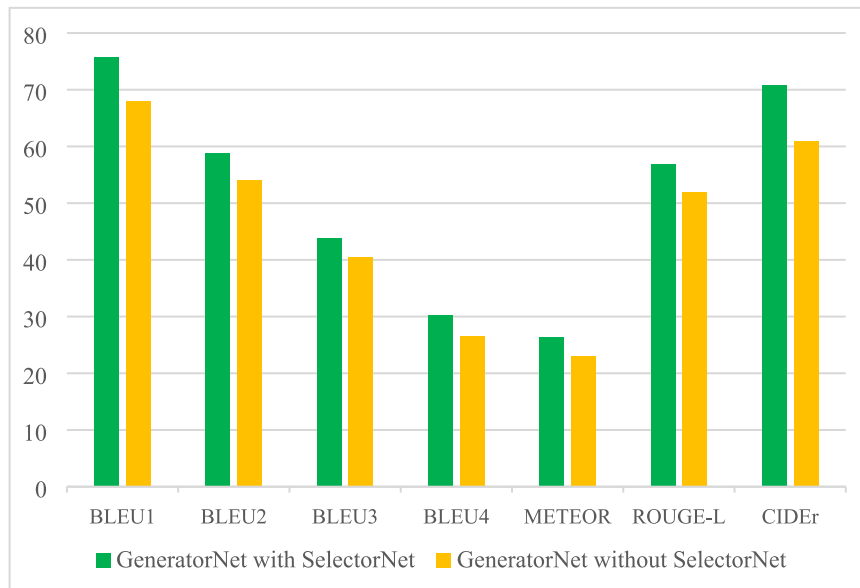


Fig. 14. Effect of the selector module on the Flickr dataset.

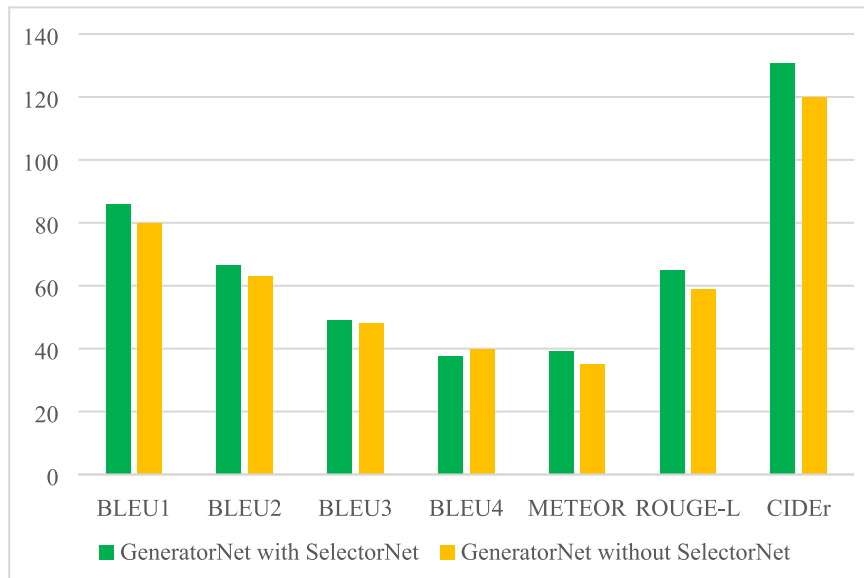


Fig. 15. Effect of the selector module on the COCO dataset.

Despite the success that the proposed transformer-based approach has achieved, the region features may fail to cover all objects in the image, although the proposed AM exploits the region-level relation, it still lacks guidance from image-level information. It makes the issue of small objects missing and incorrect relation recognition among objects, as shown in Fig. 22. Hence, similar to other models, our method fails to comprehend the primary visual contents under the noisy background, i. e., holding an umbrella in case (a) and video game in case (b). Adding more detailed visual modeling methods such as detection can relieve such issues in the future.

5. Discussion

The aim of this section is to discuss the advantages/limitations of our method. Additional details are presented in the following.

5.1. Advantages

The main property of the original transformer-based image captioning is using a convolution-free mechanism as a feature enhancer in the encoder to prepare appropriate information for the decoder. Although our model keeps this property, it develops the standard SA by using various modules in terms of caption generation. Therefore, unlike LSTMs the input data need not be processed sequentially, enabling for parallelization.

A primary advantage of our method is helping the SA to attend on the most appropriate information. In other words, the proposed eliminator module filters out some irrelevant information in the object features. Additionally, this property enhances the noise robustness of the model to handle noisy data. Moreover, the proposed MAD module provides multiple distributions to avoid a single shallow attention distribution and considers deeper internal relationships.

Furthermore, the proposed model is not influenced by irrelevant

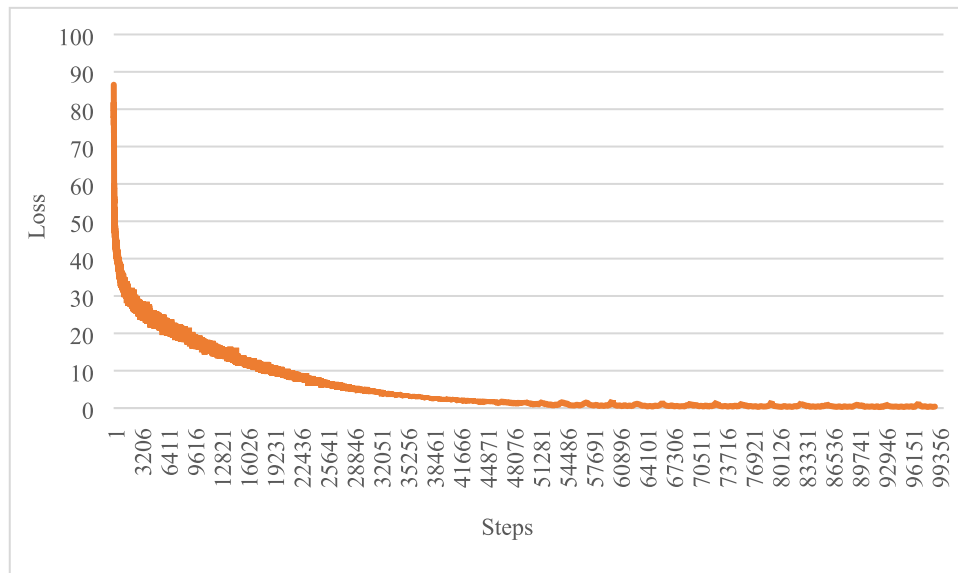


Fig. 16. Convergence of the generator network over the Flickr30k dataset.

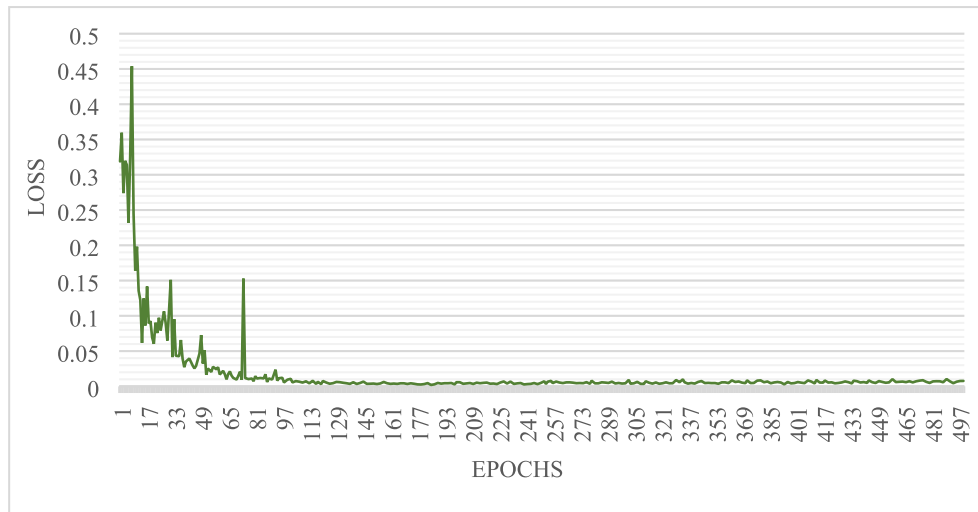


Fig. 17. Convergence of the selector network over the Flickr30k dataset.

information and thus it is more efficient than the original SA. This property causes the proposed model to accurately consider deeper internal relationships. The proposed selector module learns the similarity between sentences and images, and can assess the correctness of extracted captions, whilst the previous models rely on the greedy search to select the next best word.

Most existing SA-based modules assign attention weights to all candidate vectors based on the assumption that all vectors are relevant. It may generate some misleading or unrelated information. To address this issue, the proposed method directly constrains the attention weights. In each iteration, the hybrid attention distribution boosts more complex relationships between key and query vectors, and avoids a single shallow attention distribution.

5.2. Limitations

The ability provided by the proposed model starts to open up the possibilities for multimodal learning, where applications will be able to utilize multiple types of information, combining media such as video and images, with text.

However, similar to any transformer method, the proposed model generally incurs a high compute cost when employed on high-dimensional data, because SA has memory complexities and quadratic time with respect to the input sequence length. Although combining the local guidance with the global policy leads to enhancing the discrimination ability of our model in overlapped regions, it has computational over-head, which reduces its applicability to big data analysis. Hence, one can present a computational efficient transformer model to include inductive biases alongside transformers to improve their efficiency.

The same as with other transformer-based models, the processing performance of short sequences has not been considered, although the proposed model being able to learn long sequences efficiently. To this end, a dual structure model can be designed to combine the LSTM structure with the transformer model to generate more accurate captions.

Despite the success that the proposed transformer-based approach has reached, in the encoding step, the region features may fail to cover all objects in the image, although the AM models the region-level relation, they still lack guidance from image-level information. It makes the issue of small objects missing and incorrect relation recognition among

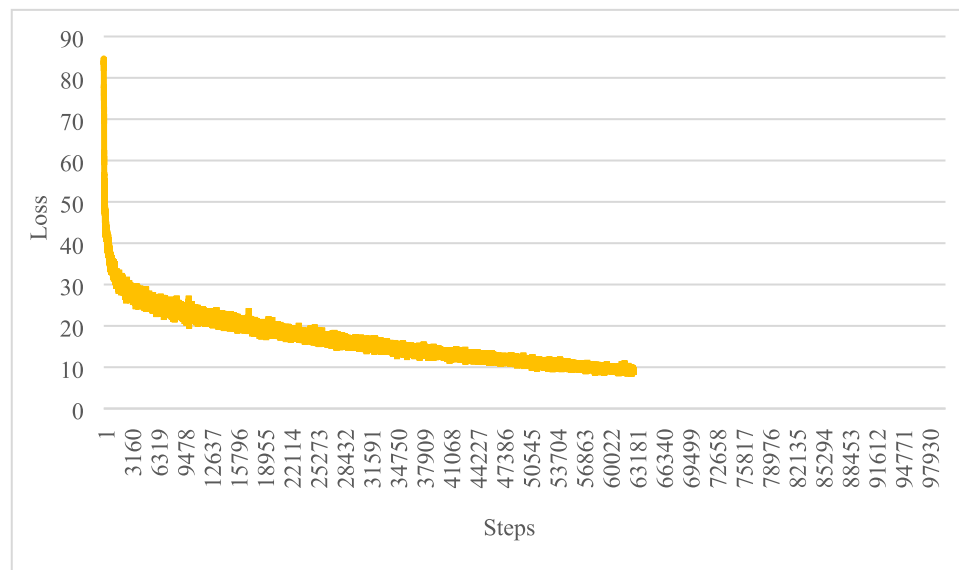


Fig. 18. Convergence of the generator network over the COCO dataset.

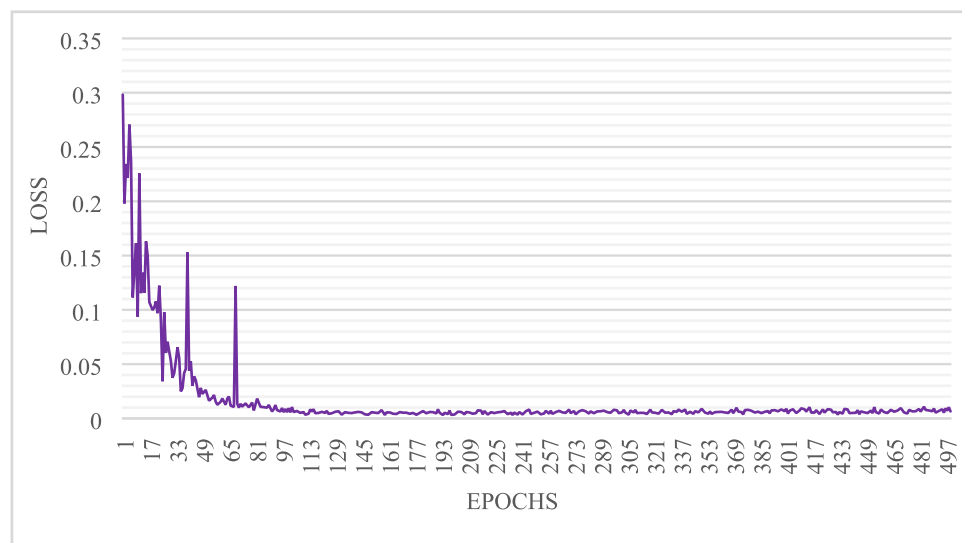


Fig. 19. Convergence of the selector network over the COCO dataset.

Table 3
Statistical student's test results.

Methods	<i>P</i> values on the Flickr	<i>P</i> values on the COCO
VIS_SAS	0.0023	0.0109
NeuralTalk	0.0002	0.0024
inject-tag	0.0007	0.0226
Based-generation	0.0027	0.0065
DSEN	0.0019	0.0048
AVTMN	0.0233	0.0166
SEM	0.0310	0.0140
EnsCaption	0.0842*	0.0418
LW + DS + Fusion	0.0001	0.0029
SCST	0.0237	0.0387
Sen	0.0181	0.0217
PAGNet	0.0389	0.0048
GAT	0.0378	0.0578 *
TransKG	0.0247	0.0153
ALT	0.0465	0.0435

objects.

6. Conclusion

- In this paper, we introduce the new transformer-based local–global strategy for effective attention alignment for image captioning. We consider an issue in existing models, which is the caption generation without recurrence architecture using word-level and sentence-level policies. To this aim, we present a transformer-based generator network to generate next words in each state, and a selector network to compute the similarity between the partial captions and visual features to select the next best word.
- The proposed transformer-based generator network improves the standard Self-Attention (SA) by using various modules, including the Mask Attention Weight (MAW) module, Self-Eliminator Module (SEG) and Multiple Attention Distribution (MAD) module in terms of caption generation. The MAW helps the SA to attend on the most appropriate information. The SEM filters out some irrelevant information in the object features. Moreover, the MAD provides multiple

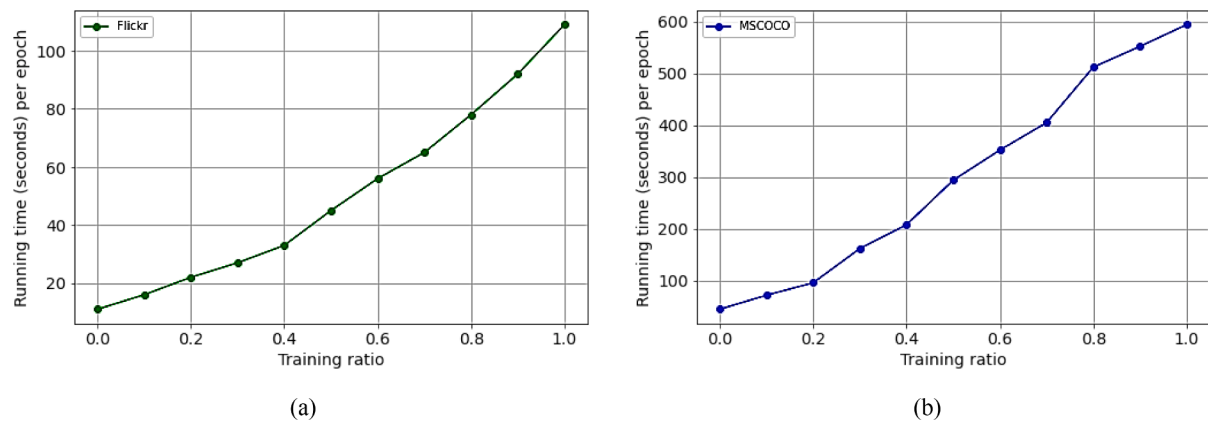


Fig. 20. The scalability of the proposed model over (a) Flickr, and (b) MSCOCO datasets.



Fig. 21. Qualitative analysis of our algorithm on the MSCOCO and Flickr datasets.

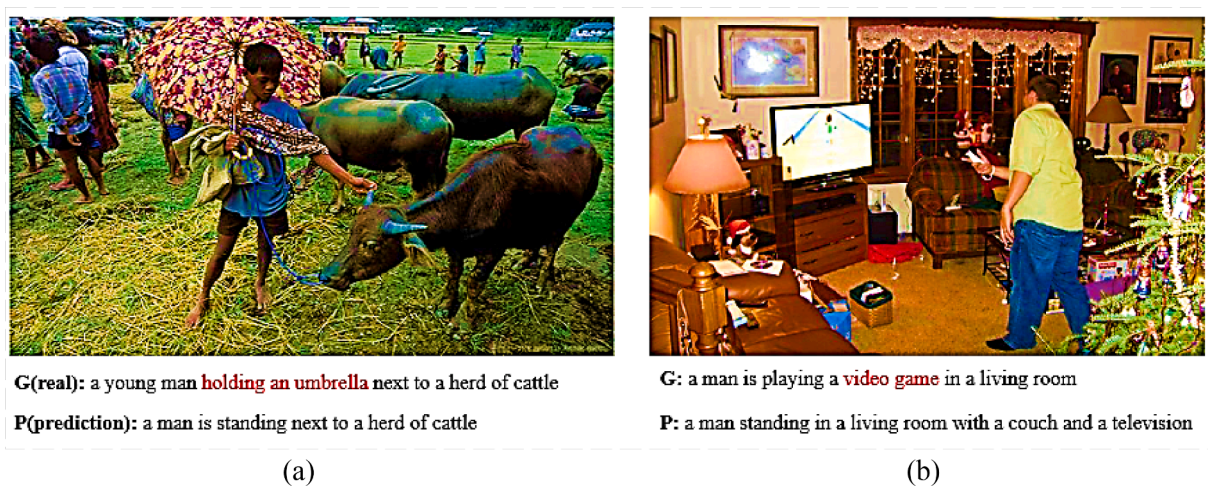


Fig. 22. Qualitative unsuccessful examples of captions predicted by the proposed model.

distributions to avoid a single shallow attention distribution and considers deeper internal relationships.

- Besides, the proposed selector network learns the similarity between sentences and images, and can measure the correctness of extracted captions. Therefore, our selector network assigns fitness scores to the partial caption through the embedding space and utilizes a global policy to optimize image captioning.
- As a result, our approach predicts the best word by employing local and global policies. With these innovations, our architecture performs better than current state-of-the-art frameworks over MSCOCO and Flickr30k datasets. Our evaluations prove that our framework consistently achieves considerable performance over state-of-the-art systems for caption generation. Finally, we aim to develop the application of our model in the field of question-answering systems in future research.

CRedit authorship contribution statement

Hashem Parvin: Methodology, Writing – original draft, Visualization. **Ahmad Reza Naghsh-Nilchi:** Conceptualization, Methodology, Validation, Investigation, Writing – review & editing. **Hossein Mahvash Mohammadi:** Conceptualization, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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